

PROBLEM SHEET 4: SOLUTIONS

1. (a) The homogeneous system is

$$\frac{du}{dt} = u(1 - u - v), \quad (1a)$$

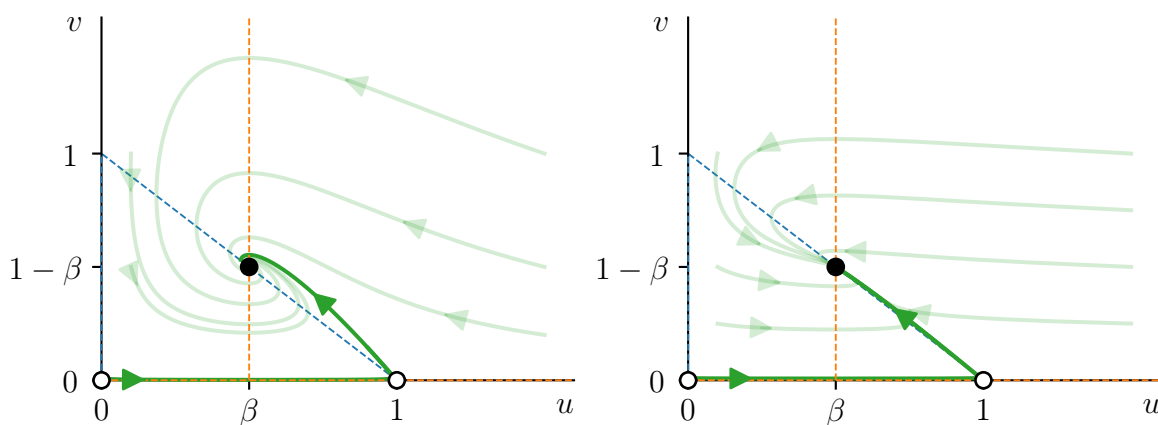
$$\frac{dv}{dt} = \alpha v(u - \beta), \quad (1b)$$

with nullclines

$$\frac{du}{dt} = 0 \implies u = 0, \quad v = 1 - u \quad (2a)$$

$$\frac{dv}{dt} = 0 \implies u = \beta, \quad v = 0. \quad (2b)$$

The phase portrait looks like:



where the case with $(\beta, 1 - \beta)$ as a stable spiral is on the left, and the case with $(\beta, 1 - \beta)$ as a stable node is on the right. Note on the right that the trajectory from $(1, 0)$ to $(\beta, 1 - \beta)$ does not go directly along the nullcline: if you zoom in, you can see it is slightly to the right (which makes sense, given that trajectories cross the nullcline vertically).

- (b) Since $(0, 0)$ is a saddle point, there is only one trajectory leaving it with $u > 0$ and we can see that it corresponds to the horizontal nullcline, i.e. trajectories leave $(0, 0)$ and go to $(1, 0)$. Therefore, there is no direct trajectory from $(0, 0)$ to $(\beta, 1 - \beta)$.
2. (a) If the stable equilibrium is at $x = \infty$, we would expect this region to expand and therefore the wave to travel *leftwards*, i.e. $z = x + ct$ with $c > 0$.
- (b) We look for solutions in z where

$$u(x, t) = u(z), \quad v(x, t) = v(z), \quad z = x - ct. \quad (3)$$

Substituting z into (🎅🌲), we obtain

$$-c \frac{du}{dz} = u(1 - u - v), \quad (4a)$$

$$-c \frac{dv}{dz} = \frac{d^2v}{dz^2} + \alpha v(u - \beta). \quad (4b)$$

Introducing $w = dv/dz$ and rearranging, we arrive at the three-dimensional system

$$\frac{du}{dz} = \frac{u}{c}(u + v - 1), \quad (5a)$$

$$\frac{dv}{dz} = w, \quad (5b)$$

$$\frac{dw}{dz} = \alpha v(\beta - u) - cw. \quad (5c)$$

The boundary conditions are

$$(u, v, w) \rightarrow (\beta, 1 - \beta, 0) \text{ as } z \rightarrow -\infty, \quad (6)$$

$$(u, v, w) \rightarrow (1, 0, 0) \text{ as } z \rightarrow \infty. \quad (7)$$

3. (a) The Jacobian of the system (  ) is

$$\mathbf{J} = \begin{pmatrix} (2u + v - 1)/c & u/c & 0 \\ 0 & 0 & 1 \\ -\alpha v & \alpha(\beta - u) & -c \end{pmatrix}. \quad (8)$$

- (b) The Jacobian, evaluated at $(1, 0, 0)$ gives us

$$\mathbf{J} = \begin{pmatrix} 1/c & 1/c & 0 \\ 0 & 0 & 1 \\ 0 & \alpha(\beta - 1) & -c \end{pmatrix}. \quad (9)$$

This matrix leads to the eigenvalue equation

$$(\lambda - 1/c)[\lambda(c + \lambda) - \alpha(\beta - 1)] = 0, \quad (10)$$

and hence the eigenvalues are

$$\lambda = \frac{1}{c}, \quad \lambda = \frac{1}{2} \left(-c \pm \sqrt{c^2 + 4\alpha(\beta - 1)} \right). \quad (11)$$

- (c) The first eigenvalue is positive so we expect an unstable solution. But what type of unstable solution? From the second expression, we see that the solution will be oscillatory (a spiral) if $c^2 < 4\alpha(1 - \beta)$. Since v cannot be less than zero, these oscillatory solutions can be discarded and this gives us a constraint on the wavespeed, just like it did with the FK equation. Namely, the permissible wave speeds satisfy

$$c^2 \geq 4\alpha(1 - \beta). \quad (12)$$

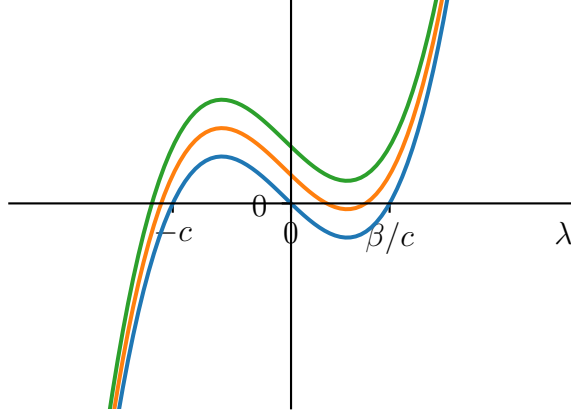
4. (a) The Jacobian, evaluated at $(\beta, 1 - \beta, 0)$ is

$$\mathbf{J} = \begin{pmatrix} \beta/c & \beta/c & 0 \\ 0 & 0 & 1 \\ \alpha(\beta - 1) & 0 & -c \end{pmatrix}, \quad (13)$$

which has eigenvalue equation

$$\lambda^3 + \lambda^2 \left(c - \frac{\beta}{c} \right) - \beta\lambda + \frac{\beta}{c}\alpha(1 - \beta) = 0. \quad (14)$$

(b) The cubic looks like



for three different values of α . (You do not need to find the points where it crosses the x -axis at this stage.)

Differentiating and setting to zero, we find

$$3\lambda^2 + 2\lambda \left(c - \frac{\beta}{c} \right) - \beta = 0, \quad (15)$$

and solving the quadratic we find

$$\lambda_{\max}, \lambda_{\min} = -\frac{1}{3} \left[c - \frac{\beta}{c} \pm \sqrt{\left(c - \frac{\beta}{c} \right)^2 + 3\beta} \right]. \quad (16)$$

(c) I am convinced. Looking at eq. (14), changing α is like doing ' $f(x) + a$ ', translating the function vertically. (See picture above.)

(d) Equation (🕯️) with $\alpha = 0$ has solutions $\lambda = 0$ and

$$\lambda^2 + \lambda \left(c - \frac{\beta}{c} \right) - \beta = 0 \quad (17)$$

$$\implies (\lambda + c) \left(\lambda - \frac{\beta}{c} \right) = 0 \implies \lambda = -c, \frac{\beta}{c}. \quad (18)$$

The eigenvalues are therefore $\lambda = 0, -c, \beta/c$.

(e) Previously, we discarded these solutions on the grounds that they would yield negative values of a population size. In those instances, however, we were dealing with equilibria where one population size was zero. Here, we have $(\beta, 1 - \beta)$, so the solution may safely oscillate without ever becoming negative. Thus, we can permit oscillatory wavefronts. While this is a fundamental difference with the FK equation where oscillations were not possible, like the FK equation the wave speed is given by the minimum possible value, namely, $c^2 = 4\alpha(1 - \beta)$.